Molek-30M: A Compute-Efficient From-Scratch MoFlow Replica for Agricultural Molecular Generation

PinoBio-GenD research note · MOLEK-001 · ZINC250k generation + distributional benchmark

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Abstract

We present **Molek-30M**, a from-scratch replica of the MoFlow invertible-flow molecular graph generator, scaled to **29,049,444 parameters** (≤30M budget) and trained on a single free-tier GPU (Kaggle T4). On the ZINC250k benchmark, Molek-30M **matches the full reference MoFlow on the corrected generation metrics** — corrected validity **99.95%**, novelty **100%**, uniqueness **99.99%** — and reaches a Fréchet ChemNet Distance of **FCD 23.40**, ~12% worse than the published MoFlow reference (20.93; lower is better) and within our pre-declared ≤1.5× tolerance. Raw validity without correction is **0%**, versus **63.11%** published for MoFlow; this gap is unexplained and under investigation, so faithfulness to the reference is not yet established. We report every metric with and without validity correction, contextualise FCD against the published ZINC250k leaderboard, and state the honest finding that MoFlow-family models — Molek included — are mid-pack on distributional fidelity, where diffusion models lead. Molek-30M is the base for an agricultural-conditioning head (future work) for Philippine crop-protection chemistry.

1. Research question

Can a MoFlow-class molecular generator, replicated **from scratch** under severe compute constraints (single free T4, ≤30M params), **faithfully reproduce the reference model's generation quality** on ZINC250k, and provide a sound base for agricultural property- conditioned generation?

2. Hypotheses

Main. A compute-constrained from-scratch MoFlow replica (≤30M params, single free T4) reproduces the reference MoFlow's ZINC250k generation quality within measurement noise.

H1 (generation quality). Corrected validity, novelty, and uniqueness each within **±1 percentage point** of the reference MoFlow (≈100% / 100% / ~100%).

H2 (distributional fidelity). Despite ~25× fewer parameters, FCD stays the same order as the reference (20.93) — specifically **≤ 1.5× the reference FCD** — not catastrophic.

H3 (correction dependence). Uncorrected (raw) validity is near-zero, reflecting MoFlow's architectural reliance on valency correction; corrected metrics are the capability measure.

3. Related work

MoFlow (Zang & Wang, KDD 2020) — invertible flow over molecular graphs with post-hoc valency correction (our reference). GraphAF (Shi 2020), GraphDF (Luo 2021) — autoregressive flows. GDSS (Jo 2022), DiGress (Vignac 2023), GraphArm (Kong 2023), CDGS (2023) — diffusion/score-based, current distributional-fidelity leaders. MOSES (Polykovskiy et al.) and FCD (Preuer et al. 2018) — evaluation standards inherited here.

4. Dataset

ZINC250k (MOSES/`molecularsets`, MIT; ZINC free): 249,455 molecules; reference split (`valid_idx_zinc.json`) 224,568 train / 24,887 test. Provenance in `data/manifests/moses-zinc250k.jsonl` (sha256, row-count G7-verified; mirror used after the DeepChem S3 403). License in `configs/licenses.yaml` (G1).

5. Methodology

Reference MoFlow, scaled (M-MOFLOW-1): `b_hidden_ch 256,256`, `a_n_flow 19`, `a_hidden_gnn 128`, `a_hidden_lin 256,64`, `b_n_flow 5`, `b_conv_lu 2`, `noise_scale 0.6` → 29,049,444 params (hard-fail > 30M). Training: AMP fp16, Adam lr 1e-3, seed 20261023, 200-epoch target; recovered checkpoint at epoch 197/200. Compute: single Kaggle T4×2; py3.13 patches M-MOFLOW-5/6/9/10. Eval: frozen reference `generate.py` (G3/G11), 10,000 samples × 5 experiments, temperature 0.85. Dual-report (M-MOFLOW-4): every metric with and without valency correction.

6. Baselines

Published ZINC250k numbers (GLAD/Kalousis 2024; GDSS/CDGS tables):

Validity is **without correction** for every row (like-for-like). GraphAF+FC, GDSS and MoFlow rows: GDSS (Jo et al. 2022) Table 2; GraphArm and DiGress: as previously listed, not re-checked.

Model	Validity w/o corr. (%) ↑	FCD ↓	Family
GraphAF+FC	68.47	16.02	autoregressive flow
GDSS	97.01	14.66	diffusion (best FCD)
GraphArm	88.23	16.26	autoregressive diffusion
MoFlow (reference)	63.11	20.93	normalizing flow
Molek-30M (ours)	**0.00 (under investigation)**	**23.40**	normalizing flow (replica)
DiGress	91.02	23.06	discrete diffusion

Molek's corrected validity (99.95%) is not comparable to this column.

7. Experiments

Each follows the `EXPERIMENT_PROTOCOL` field set.

EXP-MOLEK-001 — from-scratch training

- **Research question:** can MoFlow be trained from scratch at ≤30M on a single T4?
- **Hypothesis:** Main. **Model:** Molek-30M (29,049,444). **Dataset:** ZINC250k (pinned). **Tokenizer:** n/a (graph). **Method:** flow MLE, AMP fp16, Adam lr 1e-3, 200-ep target.
- **Hyperparameters:** b_hidden_ch 256,256 · a_n_flow 19 · a_hidden_gnn 128 · a_hidden_lin 256,64 · b_n_flow 5 · b_conv_lu 2 · noise_scale 0.6.
- **Hardware:** Kaggle T4×2. **Seed:** 20261023. **Baseline:** reference MoFlow.
- **Expected:** stable training to ~200 ep. **Actual:** epoch 197/200; ckpt sha256 `7f8374df3cb8…`; preflight 29,049,444 ≤ 30M.
- **Conclusion:** feasible (Main supported). **Next:** EXP-MOLEK-002.

EXP-MOLEK-002 — generation quality (dual-report, M-MOFLOW-4)

- **RQ:** does the replica match reference validity/novelty/uniqueness? **Hypothesis:** H1, H3.
- **Model:** Molek-30M @ ep197. **Params:** 10,000 × 5, temp 0.85. **Seed:** 20261023. **Baseline:** reference MoFlow (100/100/~100). **Suite:** frozen reference `generate.py`.
- **Expected (H1):** within ±1 pp of reference. **Actual:**

Metric	Corrected	Uncorrected	Reference	Δ vs ref
Validity	99.95% (sd 0.02)	0.00% (sd 0)	100.00%	−0.05
Novelty	100.00% (sd 0.00)	0.00%	100.00%	0
Uniqueness	99.99%	0.00%	~100%	~0
Abs. novelty	99.94%	0.00%	—	—
Abs. uniqueness	99.94%	0.00%	—	—

Per-experiment valid /10,000: 9996, 9998, 9994, 9990, 9995. - **Conclusion:** H1 supported (within ±1 pp; these corrected metrics saturate near 100%); H3 not supported as stated (published MoFlow validity without correction is 63.11%; the 0% is unexplained). **Next:** EXP-MOLEK-003.

EXP-MOLEK-003 — distributional fidelity (FCD / SNN / IntDiv)

- **RQ:** does the replica reproduce MoFlow's distribution quality? **Hypothesis:** H2.
- **Model:** Molek-30M @ ep197. **Dataset:** ZINC250k test (24,887). **Params:** 30,000 generations, temp 0.85; FCD via fcd_torch (= MOSES FCD), IntDiv/SNN via RDKit. **Seed:** 20261023. **Baseline:** published (MoFlow 20.93; GDSS 14.66; GraphAF 16.02; DiGress 23.06).
- **Expected (H2):** FCD ≤ 1.5× reference (≤ 31.4). **Actual:** FCD **23.40** (23.3965) · SNN **0.196** · IntDiv **0.908** · IntDiv2 **0.900**; FCD ratio 23.40/20.93 = **1.12**.
- **Conclusion:** H2 supported under the ≤1.5× tolerance (ratio 1.12), though ~12% worse than the reference on FCD (lower is better); both mid-pack (diffusion leads). **Next:** farm-knob conditioning; diffusion arch for FCD frontier.

Key findings

1. Molek-30M matches reference MoFlow on corrected validity/novelty/uniqueness (H1); FCD is within the ≤1.5× tolerance but ~12% worse (H2).
2. Raw validity is 0% vs 63.11% published for MoFlow (H3 not supported); unexplained, under investigation.
3. MoFlow-family is mid-pack on distributional fidelity; diffusion is the frontier lever.

Failed / negative experiments (recorded per protocol)

Training runs #1–#6 failed on sequential py3.13 incompatibilities (distutils, orderedset, rdkit.six, cairosvg) ⟶ permanent patches M-MOFLOW-5/6/9/10. D3 uncorrected eval first crashed (`UnboundLocalError` in reference `check_novelty` on 0 valid) ⟶ M-MOFLOW-13. The 2020 `moses` package is py3.13-incompatible ⟶ pivoted to `fcd_torch` for FCD (M-MOSES-1). FCD first mis-scaled vs MOSES, self-corrected against the ZINC250k baseline (F-0026).

8. Ablations

Formal ablations (temperature sweep, parameter-budget sweep, AMP on/off) are **not yet run** — stated honestly, not fabricated. Only validity correction on/off is varied here (§7, EXP-MOLEK-002). Planned: temperature ∈ {0.7, 0.85, 1.0} vs FCD/diversity; 13M vs 30M.

9. Statistical analysis

Means ± sd over 5 independent generation experiments (seed 20261023): validity sd 0.02%, novelty sd 0.00%. FCD is a single distributional estimate over 30,000 generations vs 24,887 test molecules. Bootstrap CI + Wilcoxon tooling (`benchmarks/validation/metrics.py`) is in place for downstream binder tasks.

10. Limitations

Mid-pack distributional fidelity (FCD 23.40; MoFlow-class ceiling — diffusion is the lever, not a Molek tune). Uncorrected validity 0% (vs 63.11% published for MoFlow; unexplained, under investigation). Novelty uses the reference's own SMILES-match (apples-to-apples with reference; a rigorous "true novelty" needs a canonicalisation audit). Not agri-conditioned yet (farm-knob spec'd, not trained). No wet-lab validation (none claimed). FCD absolute values vary by implementation (same ballpark, not identical-protocol).

11. Ethics

Generative chemistry for **agriculture** (crop protection), not human therapeutics. No efficacy/safety claims; wet-lab performance out of scope until benchmarked. Licenses verified (ZINC/MOSES MIT); no PII. Open, reproducible, honestly reported (G7).

12. Reproducibility

Seed 20261023; config hash in `configs/molgen-zinc250k.yaml`; pins `moflow@3611c6372602`, `moses@dd7ed6ab38e2`. Checkpoint sha256 `7f8374df3cb8…`; eval = frozen reference `generate.py`. Code: `scripts/train_moflow.py`, `eval_moflow.py`, `eval_moses_metrics.py`; results `docs/results/molek-30m-*.{md,json}`; patches M-MOFLOW-1..13 in `patches/`.

13. Conclusion

Molek-30M is a compute-efficient MoFlow-class replica: it matches the reference on the corrected generation metrics and stays within the pre-declared FCD tolerance (~12% worse), at ~25× fewer parameters on a free GPU. Its 0% raw validity (vs 63.11% published) is unexplained, so faithfulness is not yet established. An honest baseline (not distributional SOTA) and the foundation for agricultural property-conditioned generation.

Acknowledgements and AI-use statement

Work by the PinoBio-GenD generative tier of Project Katala. Method lineage: MoFlow (Zang & Wang, KDD 2020, MIT). AI assistance was used for implementation, experiment orchestration, and drafting; all numbers are measured on pinned code and verified against source result files. No external closed-API model outputs were used for training (G2).